

Coding Exam Guideline

15-16 August 2026





Agenda

- Coding Exam Guidelines
- Scoring viewpoints
- 7-Segment details



The exam has **4** questions— Choose any **3** questions

- **Q1:** 7-Segment Control — GPIO, EXTI (*optional*)
- **Q2:** ADC + GPIO
- **Q3:** UART (TX/RX) + ADC + GPIO
- **Q4:** PWM(implementation method is flexible) + ADC + GPIO



Scoring viewpoints

Scoring viewpoint	Full Score							Remark
Algorithm development	25	25pts • Meets 100% of the requirements.	20pts • Meets 80% of the requirements.	15pts • Meets 60% of the requirements.	10pts • Meets 40% of the requirements.	5pts • Meets 20% of the requirements.	0pts • Does not meet any of the requirements.	Scoring is cumulative . Students must satisfy all criteria at the lower levels before being awarded a higher-level score.
Program structure	15	15pts • Have template	12pts • Uses appropriate file separation, e.g. main.c, .c, .h	9pts • Functions are well separated and mostly follow One Role = One Function/Single Responsibility per Function • Main is clean	6pts • Functions are separated by purpose, but some functions still handle multiple responsibilities • Main is clean	3pts • Main partly crowded • Only a few functions are used	0pts • All or almost all program logic is written only in int main()	
Software Quality	10	10pts • not miss 22 rule MISRA C	8pts • 1–3 MISRA/style issues • Comment on important points	6pts • 4–7 MISRA/style issues • Variable/function names are generally clear • Easy to understand and modify	4pts • 8 - 12 MISRA/style issues • Code is somewhat difficult to understand or modify	2pts • Poor naming/readability • Code is difficult to maintain • Multiple MISRA/style issues	0pts • No comment • Unclear naming and formatting	
Total	50							

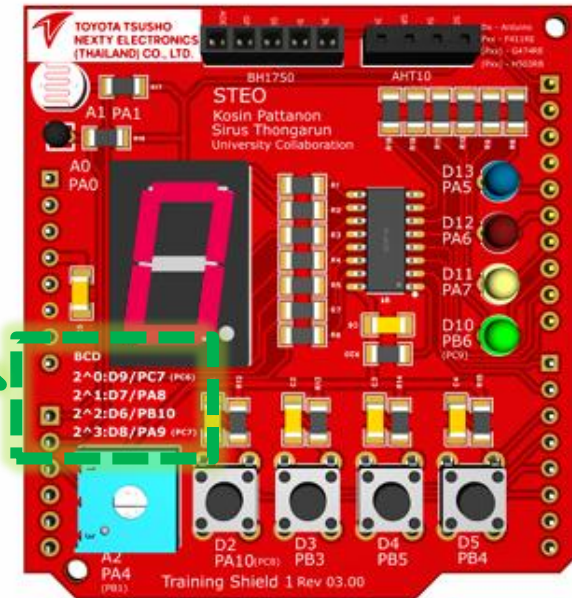
Ref:

- [Guideline_Checksheet_MISRA_C](#)
- [Template_C](#)



- The 7-Segment has the following detail:
 - STM32F411RE Pin to control 7-Segment

#	7-Segment Pin	STM32F411RE Pin
1	2^0	PC7
2	2^1	PA8
3	2^2	PB10
4	2^3	PA9



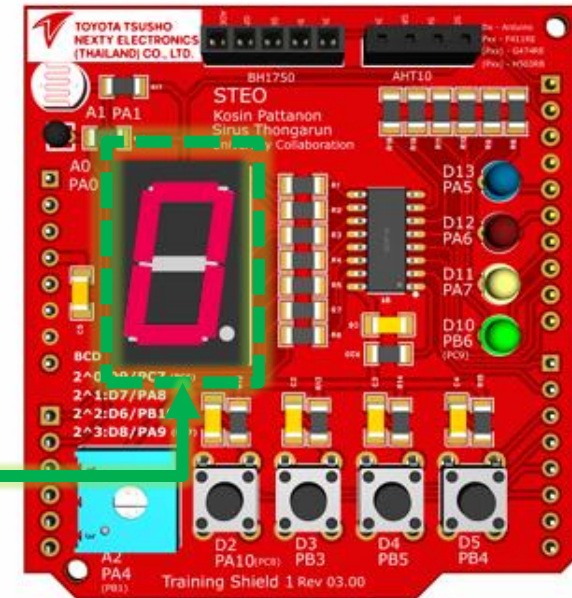
Note: If you have the blue shield, the pin labels on the silkscreen are incorrect. Please use the same pin mapping as the red shield.



7-Segment details

- The 7-Segment has the following detail:
 - Convert the number to **4-bit binary** and set each **GPIO pin** HIGH or LOW according to the table

2^3	0	0	0	0	0	0	0	1	1	1	1	1	1	1
2^2	0	0	0	0	1	1	1	1	0	0	0	0	1	1
2^1	0	0	1	1	0	0	1	1	0	0	1	1	0	0
2^0	0	1	0	1	0	1	0	1	0	1	0	1	0	1
Display Number	0	1	2	3	4	5	6	7	8	9	-	-	-	-



ありがとうございます

